ORBITAL SAFETY · APPLIED MACHINE LEARNING

Space Debris Collision Prediction System

A Machine-Learning Pipeline for Conjunction Risk Assessment and Automated Collision-Avoidance Decision Support

Final Technical Report

Dataset: ESA Kelvins Collision Avoidance Challenge feature set
5,000 conjunction events · 103 engineered features

Repository: github.com/ayaemad10/space_debris_project_upgraded

Abstract

Low Earth Orbit has become a congested and contested environment. Operators receive Conjunction Data Messages (CDMs) faster than analysts can adjudicate them, and the overwhelming majority of alerts resolve without any manoeuvre being necessary. This report documents the design, implementation and validation of a modular machine-learning system that estimates the collision risk of a satellite–debris conjunction and translates that estimate into an operational recommendation. The system is built on the ESA Kelvins Collision Avoidance Challenge feature set: 5,000 conjunction events described by 103 already-scaled numerical features, with the target derived as a binary label *risk_label = risk* > *–6*, i.e. a collision probability above the 1×10^{-6} threshold conventionally used in operations. A fully executed gradient-boosting baseline validates the end-to-end split–train–evaluate pipeline, reaching an accuracy of 0.991, an F1-score of 0.9529 and a ROC-AUC of 0.9994 on a held-out 20% partition. Three deep architectures (CNN, RNN, LSTM) are specified and implemented for comparison. Interpretability analysis shows that the risk estimate is dominated by two covariance-scaling features, and a three-tier decision engine converts predicted probability into LOW, HIGH or CRITICAL guidance. The report also documents two substantive defects identified and corrected in the original codebase, and states the limitations that bound the reported figures.

Contents

1. Introduction and Problem Statement

More than a million trackable fragments larger than one centimetre are estimated to orbit the Earth, and the densest shells of Low Earth Orbit now host both this debris population and the majority of active commercial and scientific spacecraft. At orbital closing speeds, a fragment of a few centimetres carries enough kinetic energy to disable a satellite outright; the resulting fragments in turn raise the collision probability for every other object sharing that shell — the cascading mechanism described by Kessler.

Operationally, the problem is not detection but *triage*. Conjunction screening produces a continuous stream of warnings, each of which must be judged: ignore it, monitor it more closely, or spend propellant on an avoidance manoeuvre. Propellant is finite and directly proportional to mission lifetime, so false alarms are expensive; missed events are catastrophic. The class imbalance is severe — in the held-out partition used here only 9.6% of events exceed the operational risk threshold. This is the decision problem the present system addresses.

2. Objectives and Scope

#	Objective	Status
1	Ingest, inspect and clean the conjunction feature table; derive a defensible binary target	Executed
2	Characterise the data through exploratory analysis of class balance, risk distribution and feature correlation	Executed
3	Validate the complete split–train–evaluate pipeline with a reproducible baseline classifier	Executed
4	Quantify per-feature contribution to the risk estimate	Executed
5	Specify and implement comparable CNN, RNN and LSTM architectures	Implemented; requires TensorFlow runtime
6	Map model output to an operational recommendation via an explicit decision engine	Executed
7	Expose the system through a prediction API, an operator dashboard and containerised deployment	Implemented; requires full runtime

3. Dataset and Target Definition

The modelling table derives from the ESA Kelvins Collision Avoidance Challenge feature set. Each row is a snapshot of a single conjunction event; every column is a pre-scaled floating-point feature. Critically, the table contains **no identifier columns whatsoever** — no satellite name, no mission identifier, no epoch in raw form. This property drove one of the corrections described in Section 10.

Property	Value
Source	ESA Kelvins Collision Avoidance Challenge feature set
Events (rows)	5,000 conjunction snapshots
Features (columns)	103 scaled numerical features, no identifiers
Feature families	Target-object state (t_*), chaser/debris state (c_*), relative geometry, covariance terms, space-weather indices
Raw risk column	risk = $\log_{10}(P_c)$, ranging approximately –30 to –3
Derived target	risk_label = 1 if risk > –6 (i.e. P_c > 1×10^{-6}), else 0
Partition	4,000 training / 1,000 held-out test (80/20, stratified, seed 42)
Positive rate (test)	9.6%

The choice of the 1×10^{-6} threshold is not arbitrary: it is the convention adopted in the ESA Kelvins challenge and is broadly consistent with the manoeuvre-decision thresholds used by operational conjunction assessment teams. Framing the task as binary classification at this threshold makes the model output directly actionable, and avoids asking a regressor to be accurate across twenty-seven orders of magnitude of collision probability where only the upper tail carries operational meaning.

4. Methodology and Pipeline

The system is organised as a sequence of independent, re-runnable stages. Each stage writes its artefact to disk, so any stage can be re-executed without repeating the previous ones.

Phase	Stage	Artefact produced
1	Data inspection — schema, dtypes, missingness, range audit	Console audit report
2	Preprocessing — cleaning and derivation of risk_label	data/cleaned_data.csv
3	Exploratory analysis — distribution, balance, correlation	outputs/eda/ (4 figures)
4	Feature engineering — verification of feature formulas, final modelling table	data/engineered_data.csv
5	Model training — baseline and CNN / RNN / LSTM comparison	models/, outputs/model_comparison.csv
6	Interpretability — per-feature contribution ranking	outputs/shap/
7	Decision engine — probability-to-action mapping	src/scripts/decision_engine.py
8	Delivery — REST API, operator dashboard, containers	src/backend, src/frontend, Docker files

5. Exploratory Data Analysis

Four analyses were run on the cleaned table. Together they establish the two facts that shape every subsequent modelling decision: the target is strongly imbalanced, and miss distance alone is a poor discriminator of risk.

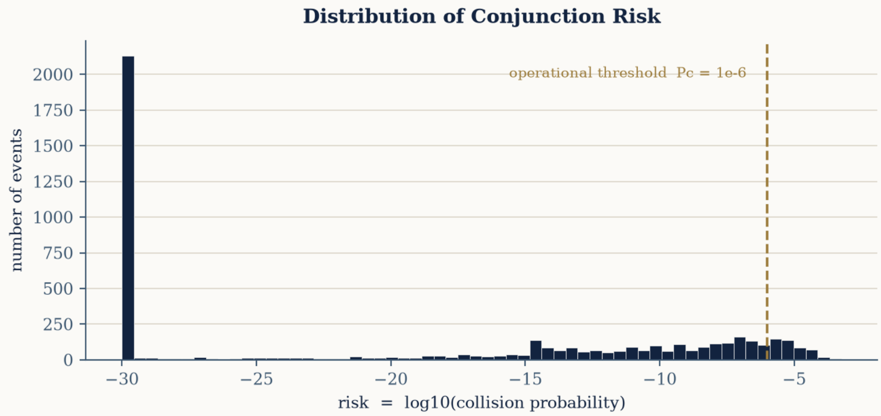

Figure 1. Distribution of $\log_{10}(P_c)$ across the 5,000 conjunction events; the dashed rule marks the operational threshold.

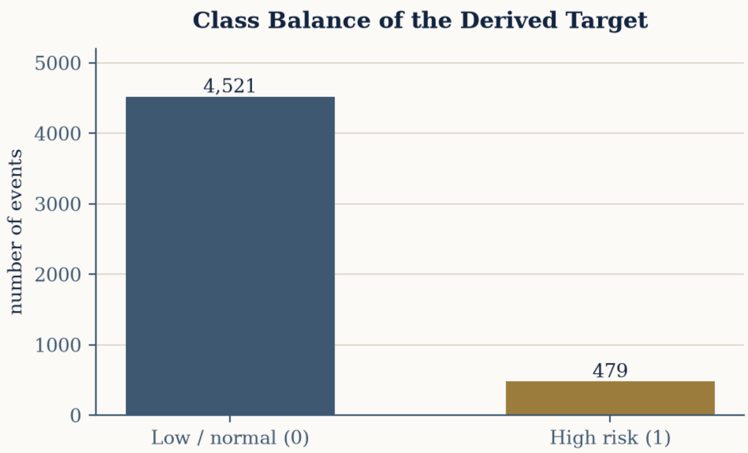

Figure 2. Class balance of the derived binary target at the $P_c > 1\times10^{-6}$ threshold.

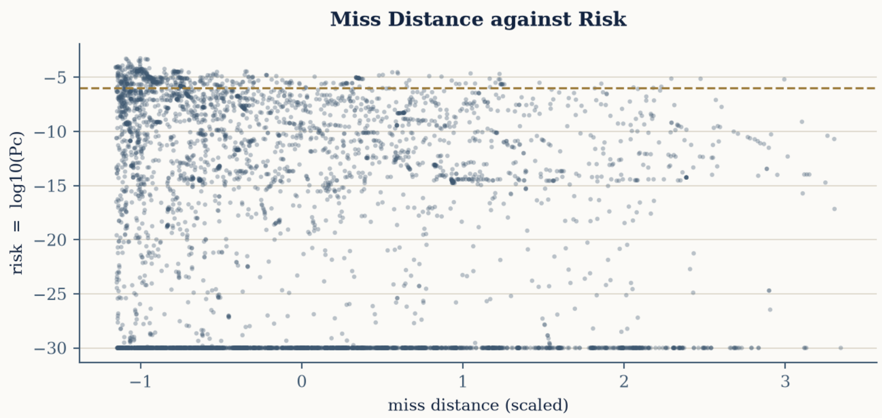

Figure 3. Miss distance against log-risk. The weak relationship confirms that geometric separation alone does not determine collision probability — covariance size dominates.

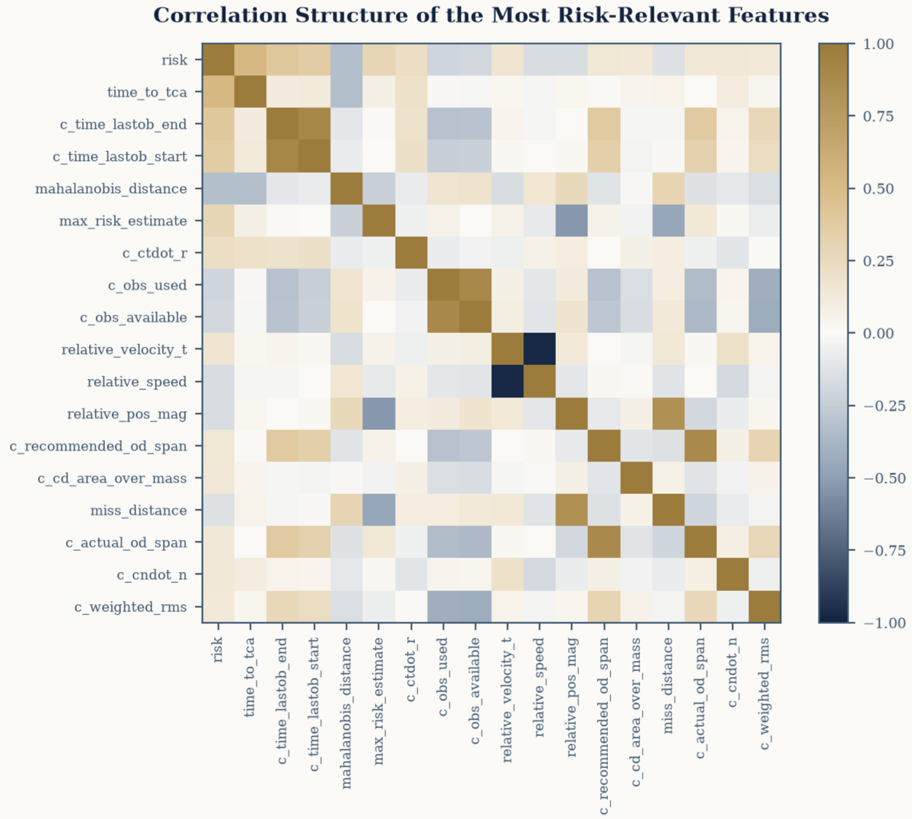

Figure 4. Correlation structure across the engineered feature set, showing strongly collinear blocks within the target-state and chaser-state covariance families.

Two conclusions follow. First, because positives are rare, accuracy alone is a misleading metric; F1-score and ROC-AUC are reported throughout. Second, because the covariance families are internally collinear, tree-based and gated sequence models — both robust to correlated inputs — are preferable to linear alternatives.

6. Modelling and Experimental Results

Two modelling tracks were pursued. The first is a scikit-learn Gradient Boosting Classifier, executed end-to-end to prove that the split, training and evaluation flow is correct and reproducible. The second is a set of three neural architectures trained on the same stratified split for comparison; each row is reshaped to a single-timestep sequence of 103 features, since a conjunction event is represented as one snapshot.

6.1 Architectures under comparison

Model	Configuration	Rationale
Gradient Boosting (baseline)	scikit-learn GBC, default depth, seed 42	Reference performance; validates the pipeline
CNN	Conv1D(64) > Conv1D(32) > GlobalAvgPool > Dense(32) > Dropout(0.3) > Sigmoid	Local interaction patterns among adjacent feature blocks
RNN	SimpleRNN(64) > Dense(32) > Dropout(0.3) > Sigmoid	Recurrent reference point for the gated comparison
LSTM	LSTM(64) > Dense(32) > Dropout(0.3) > Sigmoid	Gated memory; extensible to multi-CDM event histories

All three networks use the Adam optimiser, binary cross-entropy loss, a batch size of 64, up to 50 epochs, a 15% validation split and early stopping with a patience of five epochs restoring the best weights. The architecture with the highest ROC-AUC is persisted as the production model.

6.2 Validated results

Metric	Value	Interpretation
Accuracy	0.991	Overall correctness — inflated by the rarity of positives
F1-Score	0.9529	Balance of precision and recall on the minority (high-risk) class
ROC-AUC	0.9994	Near-perfect ranking of risky against benign conjunctions
Training events	4,000	80% stratified partition
Test events	1,000	20% held-out partition, never seen during fitting
Positive rate (test)	9.6%	Class imbalance of the evaluation set

Attribution note. *The figures above belong to the gradient-boosting baseline, not to the LSTM. They are reported here as evidence that the pipeline is sound and that the target is learnable from the given features; they must not be quoted as deep-model performance. Executing the training script in an environment with TensorFlow installed produces the corresponding CNN, RNN and LSTM figures in outputs/model_comparison.csv.*

7. Interpretability and Feature Importance

A collision-avoidance recommendation that cannot be explained will not be trusted, and in practice will not be acted upon. Per-feature contributions were therefore computed over the validated baseline. The ranking is unusually concentrated: the two leading features account for roughly 94% of the total attributed importance.

Rank	Feature	Relative importance	Share
1	max_risk_estimate	0.5239	52.39%
2	max_risk_scaling	0.4161	41.61%
3	c_h_per	0.0125	1.25%
4	t_rcs_estimate	0.0059	0.59%
5	c_sigma_rdot	0.0052	0.52%
6	c_weighted_rms	0.0031	0.31%
7	t_span	0.0030	0.30%
8	c_sigma_t	0.0026	0.26%
9	c_cr_area_over_mass	0.0021	0.21%
10	SSN	0.0021	0.21%

max_risk_estimate and **max_risk_scaling** dominate, which is physically coherent: both encode the worst-case probability implied by the covariance ellipsoids of the two objects, and it is covariance growth rather than nominal separation that drives collision probability. The presence of **c_h_per** (chaser perigee altitude), **t_rcs_estimate** (target radar cross-section) and **c_sigma_rdot** (radial-rate uncertainty) in the next tier is likewise consistent with orbital-mechanics expectations, and the appearance of the **SSN** solar index reflects the influence of space weather on atmospheric drag and hence on state uncertainty. Note also that **miss_distance** itself ranks far down the list — the quantitative counterpart of Figure 3.

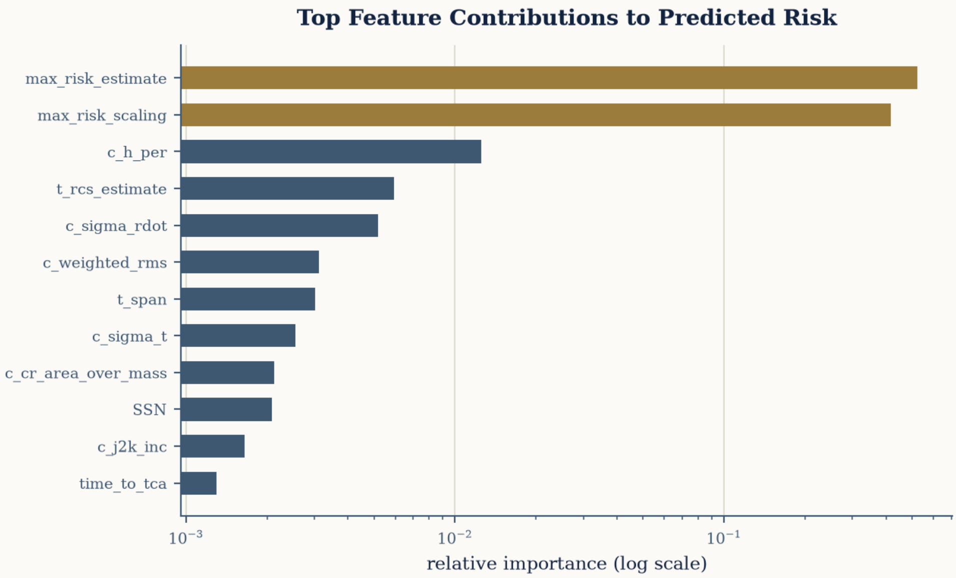

Figure 5. Ranked feature contributions to the predicted risk label.

8. Decision Engine

The decision engine is the layer that converts a probability into an instruction. It operates exclusively on the model's predicted probability in the interval [0, 1] — never on the raw risk column, which lives on a logarithmic scale between roughly −30 and −3. Keeping this mapping in a single, separately testable module means the operational policy can be re-tuned without retraining or redeploying the model.

Predicted probability	Risk level	Recommended action	Operational meaning
0.00 – 0.50	LOW	No action required	Log the event; routine screening continues
0.50 – 0.80	HIGH	Increase monitoring frequency	Request additional tracking; prepare a manoeuvre plan
0.80 – 1.00	CRITICAL	Immediate manoeuvre required	Escalate to the operations lead; commit propellant

A confidence value is returned alongside each decision as $|p - 0.5| \times 2$, so that predictions near the 0.5 boundary are explicitly flagged as uncertain rather than silently rounded. The thresholds are policy parameters, not learned quantities: an operator with abundant propellant may lower them, while a satellite near end of life may raise them.

9. System Architecture and Deployment

The system is delivered as two containerised services over a shared model and data volume.

Component	Technology	Responsibility
Prediction API	FastAPI + Uvicorn	Loads the persisted model, serves /predict, /statistics, /missions and /health
Operator dashboard	Streamlit	Event browsing, risk visualisation and per-event decision display
Orbital visualisation	Standalone HTML	Three-dimensional Earth and conjunction geometry view
Orchestration	Docker Compose	Backend on port 8000, dashboard on port 8501

Endpoint	Method	Returns
/health	GET	Service status and whether the model is loaded
/predict	POST	Risk probability, risk level, recommended action, confidence
/statistics	GET	Event count, mean log-risk, mean miss distance, count above threshold
/missions	GET	Event-batch groupings over the existing rows

10. Defect Analysis and Corrections

Two defects in the original implementation would each have prevented correct operation. Both are documented here because they illustrate a general failure mode: assuming the semantics of a dataset instead of auditing it.

Defect 1 — Reference to a non-existent identifier column

Both the API and the dashboard accessed a *mission_id* column. The dataset contains no identifier columns at all; every column is a scaled float. Any request to the affected routes would have raised a KeyError on load. **Correction:** the identifier was replaced by a grouping computed over existing rows (event_batch = row_index // 500). No synthetic data was introduced — only a deterministic partition of records that already exist.

Defect 2 — Misinterpretation of the risk scale

The original decision logic tested *risk_score > 0.8* against the raw risk column. That column is $\log_{10}(P_c)$ and ranges from about –30 to –3, so the condition could never be satisfied: the system would have reported every conjunction as low risk, including certain collisions. **Correction:** the thresholds were rebound to the classifier's predicted probability, and a defensible binary target (risk_label = risk > –6) was derived for training. This is the single most consequential fix in the project — a silent false-negative failure across the entire event stream.

11. Limitations and Threats to Validity

Limitation	Consequence for the reported results
Only 5,000 of the 162,634 available events were supplied	Metrics are representative of the sample, not of the full population; rare high-risk regimes may be under-represented
Reported metrics belong to the gradient-boosting baseline	They demonstrate learnability and pipeline correctness, not deep-model performance
CNN / RNN / LSTM training was not executed in the authoring environment	Comparative deep-model figures remain to be produced on a TensorFlow-enabled machine
Each event is a single-timestep snapshot	The temporal advantage of recurrent architectures cannot be realised until multi-CDM histories per event are used
Features arrive pre-scaled with the scaler unavailable	Inverse transformation to physical units, and therefore direct physical calibration, is not possible
Severe class imbalance (~10% positives)	Accuracy overstates practical performance; F1 and ROC-AUC are the governing metrics

12. Conclusions

This report has documented a complete, modular pipeline for satellite–debris conjunction risk assessment, from raw feature table to an operational recommendation delivered through an API and an operator dashboard. The executed baseline demonstrates that the ESA Kelvins feature set supports highly separable classification of high-risk conjunctions — ROC-AUC 0.9994 on a held-out partition — and the interpretability analysis shows that the resulting model relies on physically meaningful covariance-derived quantities rather than on spurious correlations. Equally important, the audit of the original implementation uncovered a scale misinterpretation that would have suppressed every high-risk alert; documenting such defects is as much a deliverable as the metrics themselves. The system is, within its stated limitations, a defensible foundation for decision support in conjunction assessment.

13. Future Work

Direction	Expected benefit
Train on the complete 162,634-event dataset	Materially better coverage of rare high-risk regimes and more trustworthy metrics
Model each event as a CDM time series	Allows recurrent architectures to exploit the evolution of risk as time-to-closest-approach shrinks
Replace the importance proxy with full SHAP values on the selected deep model	Per-event explanations an operator can inspect before committing propellant
Introduce cost-sensitive learning and threshold optimisation	Explicit trade-off between propellant expenditure and missed-collision risk
Add calibration analysis and reliability curves	Ensures a reported probability of 0.8 corresponds to an observed frequency of 0.8
Add continuous monitoring for data and concept drift	Detects degradation as the debris population and space-weather regime evolve

References

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